Global Optimality and Uniqueness for Nine Coulomb Charges on the Sphere: An Exact-Certificate Proof in Lean 4

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Abstract

For nine distinct points on the unit sphere in real three-space, the sum of reciprocal Euclidean distances is minimized by a particular tricapped trigonal prism. We describe a computer-assisted proof of global optimality and uniqueness up to orthogonal transformations and relabeling, together with its Lean 4 formalization. The candidate is defined by a radical stationarity equation and a rational isolating interval, not by numerical coordinates. A rational degree-20 three-point semidefinite certificate is nearly sharp, with its entire residual paid explicitly. Exact Bernstein certificates force pair inner products into narrow windows. Finite Gram-matrix classification gives one coherent relabeling, and a certified ambient Hessian bound completes the argument locally. The resulting Lean theorem has no certificate hypotheses. Its recorded axiom closure consists only of propositional extensionality, classical choice, and quotient soundness. A separately implemented kernel also accepted the exported complete theorem and its dependencies. The source development and this draft received substantial AI assistance. Independent human review and reproduction by another operator remain requested; no priority claim is made here.

1 Statement and scope

Let

$$\mathcal{C}_9 = \{(x_0, \dots, x_8) \in (\mathbb{R}^3)^9 : \|x_i\| = 1 \text{ for all } i, \quad x_i \neq x_j \text{ for } i \neq j\}.$$

The energy convention throughout is

$$E(X) = \sum_{0 \le i < j < 9} \frac{1}{\|x_i - x_j\|} = \sum_{i<j} f(\langle x_i, x_j \rangle), \qquad f(t) = \frac{1}{\sqrt{2-2t}}. \tag{1}$$

Thus there are 36 terms; neither ordered pairs nor reciprocal squared distances are used. Denote by $P = P(r_*)$ the configuration defined in Section 2.

Theorem 1.1. *For every $X \in \mathcal{C}_9$, $E(X) \ge E(P)$. Moreover,*

$$E(X) = E(P) \quad \Longleftrightarrow \quad \exists Q \in O(3),\ \exists \sigma \in S_9, \forall i, \quad x_i = Q p_{\sigma(i)}.$$

The parameter r_* exists and is uniquely determined by the exact characterization below, and $P \in \mathcal{C}_9$.

The formal statement is independent of the global proof: the root file `lean/Challenge.lean` imports the candidate definitions, not the certificate assembly. It asks for parameter existence and uniqueness, global optimality, and both directions of the equality characterization. The final theorem is

```
ThomsonN9.thomson_nine_complete:
ThomsonN9.Challenge.CompleteTarget.
```

The configuration space uses Mathlib's `EuclideanSpace` indexed by `Fin3`, with $\mathbb{R}$ as the scalar field. Congruence uses a real linear isometry equivalence and a permutation. In particular, reflections are allowed. Separate fixed-candidate theorems remove the parameter argument.

This paper follows the *selected formal route*: the reduced D60 certificate, Bernstein positivity, finite Gram-window classification, and the ambient augmented Hessian. Earlier Hermite-remainder and higher-precision certificate routes remain in the research archive, but are not substituted for the proof checked here. Large finite data are part of the accompanying artifact, not printed as thousands of coefficients in this paper. We state their mathematical obligations and identify their checkers explicitly. In the publication repository, all proof-source, certificate, and verification paths below are relative to the `proof/` directory unless explicitly identified as publication-level paths. The supplemental upstream challenge bridge is instead in the publication-level `bridge/` directory.

1.1 Relationship to prior work

Cohn and Kumar's universal-optimality theorem gives a conceptual framework for several classical spherical energy minima, including the tetrahedron, octahedron, and icosahedron [1]. Schwartz's five-electron proof combines global interval elimination with local analysis [2]. For eight points, Kryvonos, Liehr, and Taylor establish Coulomb optimality and uniqueness of an optimized square antiprism using exact certificates, with a Lean formalization [3, 4]. Tran's seven-point development uses three-point bounds, geometric localization, and local rigidity [5]. These works motivated the present architecture, especially the use of a nearly sharp global certificate followed by a quantitative local argument.

The N9 implementation does not import the seven- or eight-point optimality theorems as premises. An Apache-2.0-licensed Schoenberg dependency closure from the `delsarte` project is vendored with attribution and compatibility changes recorded [6]. The inspected upstream commits are specified in the bibliography and artifact. The separate full N8 project rebuilds were not completed in this investigation; source inspection is not presented as successful reproduction of those projects.

2 The exact candidate

Put

$$e_0 = (1, 0), \qquad e_1 = (-1/2, \sqrt{3}/2), \qquad e_2 = (-1/2, -\sqrt{3}/2).$$

For $0 < r < 1$, let $z = \sqrt{1 - r^2}$ and define

$$p_{k,+} = (re_k, z), \quad p_{k,-} = (re_k, -z), \quad c_k = (-e_k, 0), \qquad k = 0, 1, 2. \tag{2}$$

These are nine distinct unit vectors. Rotations through $2\pi/3$, reflection in the equatorial plane, and a vertical reflection preserve the configuration. They generate D_{3h}. Under this group the six prism points form one orbit and the three equatorial caps form the other. The cyclic rotation subgroup

instead has three orbits of size three. Symmetry is a property of this candidate, not an assumption on a competing configuration.

The six pair types give the following exact spectrum.

Pair type	Inner product	Multiplicity
Same upper or lower triangle	$1 - 3r^2/2$	6
Vertically opposite prism points	$2r^2 - 1$	3
Other upper/lower pairs	$r^2/2 - 1$	6
Two caps	$-1/2$	3
Cap and opposite prism direction	$-r$	6
Other cap/prism pairs	$r/2$	12

Consequently,

$$F(r) = E(P(r)) = \frac{2\sqrt{3}}{r} + \frac{3}{2\sqrt{1-r^2}} + \frac{6}{\sqrt{4-r^2}} + \sqrt{3} + \frac{6}{\sqrt{2+2r}} + \frac{12}{\sqrt{2-r}}. \tag{3}$$

Define

$$\begin{aligned} S(r) = F'(r) = &-\frac{2\sqrt{3}}{r^2} + \frac{3r}{2(1-r^2)^{3/2}} + \frac{6r}{(4-r^2)^{3/2}} \\ &- \frac{6}{(2+2r)^{3/2}} + \frac{6}{(2-r)^{3/2}}. \end{aligned} \tag{4}$$

All square roots here are positive. Let

$$A = 710548334092953811327704179272791243768 0, \qquad I = [A/10^{40}, (A+1)/10^{40}].$$

The exact definition of r_* is the unique $r \in I$ with $S(r) = 0$.

Lemma 2.1. *There is exactly one zero r_* of S in I. It is in the interior of I and is the unique minimizer of F on $(0, 1)$.*

Proof. Continuity holds on $(0, 1)$. Direct differentiation gives

$$S'(r) = \frac{4\sqrt{3}}{r^3} + \frac{3(1+2r^2)}{2(1-r^2)^{5/2}} + \frac{12(2+r^2)}{(4-r^2)^{5/2}} + \frac{18}{(2+2r)^{5/2}} + \frac{9}{(2-r)^{5/2}} > 0.$$

Exact rational interval evaluations establish $S(A/10^{40}) < 0$ and $S((A+1)/10^{40}) > 0$. These two finite evaluations, including all denominator signs and square-root enclosures, are checked in the parameter certificate modules. The intermediate value theorem and strict monotonicity give the claim. Positivity of S' also gives strict decrease of F before the root and strict increase afterwards. See `parameter_exists_unique` and `FamilyMinimum.lean` for the checked argument. □

The original radical equation is retained instead of replacing it by an unqualified polynomial equation obtained by squaring. The artifact contains a separate degree-104 integer elimination polynomial, but no irreducibility claim is needed and it is not the defining premise of the formal theorem. For orientation only,

$$\begin{aligned} r_* &\approx 0.710548334092953811327704179272791243768 0, \\ E(P) &\approx 25.759986531269831564075702273760494 0. \end{aligned}$$

Neither decimal is a proof input.

3 Global scalar certificates

For the rest of the proof set

$$M = \frac{5593}{10000}, \qquad \varepsilon = 10^{-4}, \qquad \delta = 10^{-8}.$$

An exact rational upper bound U for $E(P)$ is defined in `Certificates/ReducedEnergy.lean`:

$$U = \frac{5151997306253966312815140454752098805728382727295351889952429}{2000000000000000000000000000000000000000000000000000000000000000000000000000}.$$

It comes from interval evaluation of F at a rational radius and Lemma 2.1. Thus $E(P) \le U$ uses only minimality within the symmetric family, not Theorem 1.1.

3.1 Separation by a two-point bound

Lemma 3.1 (Separation). *If $X \in \mathcal{C}_9$ and $E(X) \le E(P)$, then $\langle x_i, x_j \rangle < M$ for all $i \ne j$.*

Certificate proof. The separation polynomial q has a Legendre expansion $q(t) = \sum_k a_k P_k(t)$, with $P_k(1) = 1$. Its nonzero coefficients are

k	0	1	2	3	7
$10^{12} a_k$	806045161636	460032960362	200888085899	63087598053	5677717462

k	12	17	22	27
$10^{12} a_k$	490951549	542566025	88576655	134439481

The scalar certificate proves $q(t) \le f(t)$ on $[-1, 1)$. Schoenberg positivity gives

$$\sum_{i<j} q(\langle x_i, x_j \rangle) \ge B, \qquad B = \frac{81 a_0 - 9 q(1)}{2}.$$

The separate tail certificate proves, for $M \le t < 1$, $f(t) - q(t) \ge T - B + 10^{-8}$, where

$$T = \frac{6439996633}{250000000}, \qquad B = \frac{25728382789209}{10^{12}}, \qquad E(P) \le T.$$

If one pair were in this tail, summing its excess together with the nonnegative excesses of all other pairs would yield $E(X) > T$, a contradiction. The literal polynomial/Legendre linkage and both positivity certificates are checked in `LP/ConcreteLinkage.lean` and `ClaudeL2b/LPAdapterCodex.lean`; the application is `sublevel_separation`. □

3.2 Three-point positivity and an explicit residual

For a triple of unit vectors write its three inner products as u, v, t. They belong to the capped Gram domain

$$\mathcal{D}_M = \{(u, v, t) \in [-1, M]^3 : g(u, v, t) := 1 + 2uvt - u^2 - v^2 - t^2 \ge 0\}.$$

The three-dimensional rooted kernels use the polynomial recurrence

$$Q_0 = 1, \quad Q_1 = t - uv, \quad Q_{k+2} = 2(t - uv) Q_{k+1} - (1 - u^2)(1 - v^2) Q_k.$$

Products with univariate features and rational rectangular Gram factors give positive moment blocks. After symmetrizing over the six edge orders, the repeated-index correction for nine points is

$$\mathcal{K} = 7\mathcal{S}(u,v,t) + \mathcal{S}(u,u,1) + \mathcal{S}(v,v,1) + \mathcal{S}(t,t,1) + \mathcal{S}(1,1,1)/8.$$

The proved positivity assertion is a nonnegative *sum* of $\mathcal{K}$ over ordered distinct triples of a configuration. Individual values of $\mathcal{K}(u,v,t)$ need not be nonnegative.

The reduced certificate supplies a degree-20 rational polynomial h and an identity on $\mathcal{D}_M$ of the form

$$\frac{h(u)+h(v)+h(t)}{3} - \frac{U}{36} = \sum_b \mathcal{K}_b(u,v,t) + \sum_s w_s(u,v,t) q_s(u,v,t)^2 + \sum_\alpha c_\alpha u^{\alpha_1} v^{\alpha_2} t^{\alpha_3}. \tag{5}$$

The weights are nonnegative rational multiples and permutations of 1, $1+u$, $M-u$, g, $(1+u)(M-u)$, and $g(1+u)(M-u)$. The factor matrices are rational, so their positive semidefiniteness follows from their factored form, without a numerical eigenvalue test.

Let C be the rational value `Claude.L1.certifiedEnergyReduced` and $L = U - C$. The coefficient bound certifies $36\sum_\alpha |c_\alpha| \le L < 10^{-46}$. Its approximate value is $2.396 \cdot 10^{-54}$; importantly, it is not zero.

Lemma 3.2 (Near-sharp pair bound)**.** *For nine unit vectors whose distinct-pair inner products are at most M,*

$$U - L \le \sum_{i<j} h(\langle x_i, x_j\rangle).$$

Proof. Sum (5) over all $9 \cdot 8 \cdot 7 = 504$ ordered distinct triples. The left side becomes

$$14\left(\sum_{i<j} h(\langle x_i, x_j\rangle) - U\right).$$

The kernel sum and weighted squares are nonnegative. Every residual monomial has absolute value at most one, so the right side is at least $-504\sum_\alpha |c_\alpha|$. Dividing by 14 gives the claimed bound. □

The coefficient identity is a finite integer verification, not a comparison at sample points. Its degree box has $21^3 = 9261$ entries and covers all $\binom{23}{3} = 1771$ monomials of total degree at most 20, including coefficients that must be zero. The final instantiation is `Claude.L1.reduced_pair_bound`. The D60 label refers to the 10^{60} rational rounding scale, not to the polynomial degree.

3.3 Bernstein minorants and exterior gaps

Write $s_1(r), \dots, s_6(r)$ for the six inner products in the candidate spectrum. The certificate establishes the following exact scalar assertions.

Lemma 3.3 (Minorant and gaps)**.** *The reduced polynomial satisfies $h(t) \le f(t)$ for $-1 \le t < 1$. For every $r \in I$:*

1. *If $-1 + \varepsilon \le t \le M$ and $|t - s_j(r)| \ge \varepsilon$ for all j, then $f(t) - h(t) \ge 10^{-46}$.*
2. *If $-3/4 \le t \le M$ and $|t - s_j(r)| \ge \delta$ for all j, then $f(t) - h(t) \ge 10^{-33}$.*

Exact finite verification. For a polynomial a and a gap γ, substitute $y = \sqrt{(1-t)/2}$; then $0 < y \le 1$, $t = 1 - 2y^2$, and $f(t) = 1/(2y)$. It suffices to prove

$$1 - 2y\big(a(1-2y^2) + \gamma\big) \ge 0.$$

For degree 20 this is a polynomial of degree at most 41. On a rational subinterval it is transformed affinely to $[0,1]$ and expanded in the Bernstein basis. Nonnegative Bernstein coefficients imply nonnegativity on that whole interval. Complete subdivision trees are checked, including their coverage; accepting a few selected leaves would not suffice.

The implemented route first proves a rational precursor minorant and its coarse gaps, then proves an exact coefficient inequality transferring them downwards to h. Refined gaps are checked with their own finite data. Rational enclosures of $s_j(r)$ enlarge the exterior intervals to be checked, so no unexamined interval between contact windows is discarded. The checked endpoints are `Claude.L2b.hNew_le_inv_dist`, `Claude.L2b.exterior_gap_new`, and `Claude.L2b.refined_gap_new`. □

For a configuration in the sublevel $E(X) \le E(P)$, Lemmas 3.1 and 3.2 imply

$$0 \le \sum_{i<j} \big(f(\langle x_i, x_j\rangle) - h(\langle x_i, x_j\rangle)\big) \le L < 10^{-46}. \tag{6}$$

Each summand is nonnegative. Hence a single exterior pair from the first gap assertion would contradict (6). Every pair therefore belongs to a contact window of radius ε or the retained interval $[-1, -1+\varepsilon]$. Exclusion of that last interval is a geometric step, not an assumption of the minorant certificate.

4 Finite Gram rigidity and refined localization

Lemma 4.1 (Seven-window rigidity)**.** *Let $r \in I$, and let $x_0, \dots, x_8$ be unit vectors in $\mathbb{R}^3$. Suppose every distinct-pair inner product lies within ε of some $s_j(r)$ or lies in $[-1, -1+\varepsilon]$. Then the latter possibility never occurs. There is a permutation σ such that*

$$|\langle x_{\sigma(i)}, x_{\sigma(j)}\rangle - \langle P(r)_i, P(r)_j\rangle| \le \varepsilon \qquad \text{for all } i, j.$$

If the original six contact windows have a common radius $0 \le \eta \le \varepsilon$, the same conclusion holds with η.

Finite classification with checked geometric soundness. The six rationally enclosed windows are pairwise disjoint, so they assign a unique color to each edge. First, for a near-antipodal pair and a third point, the two other edges lie in one of $7^2 = 49$ pairs of windows. Exact interval evaluation in each case contradicts nonnegativity of the three-point Gram determinant. Next, the six-window hypothesis forces a triple with Gram determinant greater than $1/100$. The anchor-free certificate checks the remaining colored four-point possibilities and uses the vanishing of their four-dimensional Gram determinant in $\mathbb{R}^3$.

For an invertible anchor Gram matrix G_A and another point with inner-product column v against the anchor, the unit-norm condition gives

$$v^{\mathsf{T}} \operatorname{adj}(G_A) v = \det G_A.$$

For two additional points with columns v, w and mutual inner product t,

$$v^{\mathsf{T}} \operatorname{adj}(G_A) w = t \det G_A.$$

Interval versions of these polynomial equalities reject impossible edge-color extensions. A finite coverage theorem checks all remaining anchor and extension records and shows that they have exactly the candidate's edge pattern up to one permutation. This step does not choose 36 independent nearest contacts. Its exact interval operations, geometric implications, exhaustive coverage, and reference-color identification are proved separately in the `ClaudeWindow/` modules.

For the narrower-window conclusion, uniqueness of the coarse colors ensures that the same color assigned by the classification is the one supplied by the narrow hypothesis. This transfers the smaller error without repeating the finite enumeration. □

Proposition 4.2 (Refined sublevel localization)**.** *If $X \in \mathcal{C}_9$ and $E(X) \le E(P)$, then there is one permutation σ with*

$$|\langle x_{\sigma(i)}, x_{\sigma(j)}\rangle - \langle p_i, p_j\rangle| \le 10^{-8} \quad \text{for all } i, j. \tag{7}$$

Proof. Apply Lemma 4.1 to the coarse localization obtained from (6). Its coherent pattern implies every pair inner product is at least $-3/4$: the smallest candidate contact is $r_*^2/2 - 1$, and its coarse window is above that bound. The second assertion of Lemma 3.3 now applies. Since $L < 10^{-46} < 10^{-33}$, no pair can lie outside the windows of radius δ. The narrow version of Lemma 4.1 gives the conclusion. The formal result is `sublevel_refined_gram`. □

5 Alignment and the local energy gap

Lemma 5.1 (Quantitative orthogonal alignment)**.** *Under (7), there exist $Q \in O(3)$ and the corresponding relabeling such that, with $y_i = Qx_{\sigma(i)}$, $b_i = y_i - p_i$, and $\rho^2 = \sum_i \|b_i\|^2$, one has*

$$\rho \le 10^{-4}, \qquad \sum_i \big(b_i[a]p_i[c] - b_i[c]p_i[a]\big) = 0 \quad (a, c \in \{0, 1, 2\}). \tag{8}$$

Structure of the checked estimate. The first three prism points give a nondegenerate reference frame. Constructing orthonormal frames and propagating entrywise Gram errors bounds the best orthogonal displacement by $10000\delta = 10^{-4}$. The displacement minimum over $O(3)$ exists by compactness. Varying a minimizing transformation by planar rotations gives the second identity in (8). Thus the rotation gauge is obtained for an arbitrary competitor, rather than imposed as an extra hypothesis on its coordinates. The quantitative frame inequalities are in `GramAlignment.lean` and `CandidateAlignment.lean`; the minimizing-transformation argument is in `Alignment.lean`. □

The local calculation is performed on 27 ambient coordinates, avoiding a numerically chosen basis for the 15 nonrotational tangent directions. For $d_{ij} = p_i - p_j$ put

$$B(d) = \frac{3dd^{\mathsf{T}}}{\|d\|^5} - \frac{I_3}{\|d\|^3}, \qquad \lambda_i = -\frac{1}{2}\sum_{j \ne i} \frac{1}{\|d_{ij}\|}.$$

Let H be the block Laplacian of the $B(d_{ij})$, minus the block diagonal matrix with blocks $\lambda_i I_3$. It is the constrained Hessian, with the radial Lagrange-multiplier correction included. Let R be the 27×3 matrix of infinitesimal rotation columns at P. The gauge in (8) is precisely $R^{\mathsf{T}} b = 0$.

Lemma 5.2 (Certified augmented Hessian)**.** *For the exact candidate parameter,*

$$H + RR^{\mathsf{T}} - \frac{7}{50} I_{27} \succeq 0. \tag{9}$$

Exact interval and congruence certificate. An arithmetic circuit evaluates enclosures for the entries of the matrix in (9), uniformly for the entire rational interval I, using the integer scale 2^{180}. The circuit contains 37,815 checked nodes. Separate semantic lemmas identify the circuit entries with the actual coordinates, distances, radial coefficients, pair Hessians, and rotation columns.

For the resulting symmetric matrix A, a rational unit upper triangular matrix C is supplied. The certificate evaluates $C^{\mathsf{T}}AC$ through 1,458 dot products and 729 final enclosures. Its diagonal entries are at least $1/1000$ and its off-diagonal entries have absolute value at most $1/100000$. There are 26 off-diagonal entries in each row, so the diagonal-dominance margin is

$$\frac{1}{1000} - \frac{26}{100000} = \frac{37}{50000} > 0.$$

Strict diagonal dominance proves positivity; invertibility of the unit upper triangular C transfers it to A. Denominator positivity and directions of all integer-to-rational enclosure comparisons are proved in the checking library. The final theorem is `Certificates.AmbientKernel.candidate_augmented_hessian_posSemidef`. No floating-point spectrum or condition-number estimate is a premise. □

Full candidate stationarity is proved from (4) in `LocalStationarity.lean`; it is not inferred just from a numerical gradient. In the above notation the energy gradient at p_i is $\lambda_i p_i$. Also every candidate pair distance is at least one.

Lemma 5.3 (Local coercivity)**.** *Suppose $Y \in \mathcal{C}_9$, $b_i = y_i - p_i$, $\rho^2 = \sum_i \|b_i\|^2$, $0 \le \rho \le 1/500$, and $R^{\mathsf{T}}b = 0$. Then*

$$E(Y) - E(P) \ge \frac{\rho^2}{50}. \tag{10}$$

Proof. An algebraic reciprocal-distance estimate, valid for $\|d\| \ge 1$ and $\|w\| \le \|d\|/200$, is

$$\frac{1}{\|d+w\|} \ge \frac{1}{\|d\|} - \frac{\langle d, w\rangle}{\|d\|^3} + \frac{3(\langle d, w\rangle)^2 - \|d\|^2\|w\|^2}{2\|d\|^5} - \frac{51}{50}\|w\|^3.$$

Apply this to $d = d_{ij}$ and $w = b_i - b_j$. The sphere identities $2\langle p_i, b_i\rangle = -\|b_i\|^2$ and stationarity convert the linear sum to the radial correction in H. Moreover, $\sum_{i<j}\|b_i - b_j\|^3 \le (27/2)\rho^3$. Thus the formal algebraic estimate is

$$E(Y) - E(P) \ge \frac{1}{2}b^{\mathsf{T}}Hb - \frac{1377}{100}\rho^3.$$

By Lemma 5.2 and $R^{\mathsf{T}}b = 0$, $b^{\mathsf{T}}Hb \ge (7/50)\rho^2$. Finally, $7/100 - (1377/100)\rho \ge 1/50$ for $0 \le \rho \le 1/500$. This proves (10). All analytic-to-algebraic transitions are proved in `LocalPotential.lean`, `LocalEstimate.lean`, and `LocalCoercivity.lean`; no numerical Taylor remainder is used. □

6 Global conclusion and stability

Proof of Theorem 1.1. Consider any $X \in \mathcal{C}_9$. If $E(X) > E(P)$, the lower bound and the failure of equality are immediate. Otherwise Proposition 4.2 and Lemma 5.1 give a congruent configuration Y satisfying the gauge and $\rho \le 10^{-4} < 1/500$. Congruence preserves energy. Lemma 5.3 now gives

$$0 \ge E(X) - E(P) = E(Y) - E(P) \ge \rho^2/50 \ge 0.$$

Hence $\rho = 0$, $Y = P$, and X is congruent to P. Conversely, any orthogonal image and relabeling of P has its energy. Parameter existence and admissibility were established independently in Section 2. This completes all conjuncts of the formal target without an unproved localization, positivity, or symmetry hypothesis. □

The available certificate slack also gives a quantitative statement. Define the squared distance to the candidate orbit by

$$d_{\mathrm{orb}}(X,P)^2 = \min_{Q\in O(3),\sigma\in S_9} \sum_i \|Qx_{\sigma(i)} - p_i\|^2.$$

Corollary 6.1 (Quantitative stability)**.** *If $X \in \mathcal{C}_9$ and $E(X) \le E(P) + 10^{-47}$, then*

$$d_{\mathrm{orb}}(X,P)^2 \le 50\big(E(X) - E(P)\big).$$

The checked proof repeats the slack localization with the added 10^{-47} budget and applies the same local bound. Its public theorem is `thomson_nine_quantitative_stability`. This corollary was checked in Lean, but the recorded separate-kernel export described next selected the complete optimality/uniqueness theorem, not this additional root.

7 Formalization, reproduction, and trust boundaries

Paths in the following table are under `lean/ThomsonN9/`, except the explicit root and `Claude */` entries. Names in the table are theorem or module identifiers, not additional mathematical assumptions.

Mathematical component	Formal implementation
Definition and target	`Basic.lean`, `Candidate.lean`, root `Challenge.lean`
Parameter and family	`ParameterExists.lean`, `CandidateEnergy.lean`, `FamilyMinimum.lean`
Separation and SDP	`LP/Concrete.lean`, `SDP/CertificateBound.lean`, `ClaudeL1/Main.lean`
Bernstein gaps	`Bernstein/`, `ClaudeL2b/ReducedMinorant.lean`, `ClaudeL2b/RefinedWindows.lean`
Finite Gram classification	`ClaudeWindow/Main.lean`, `ClaudeWindow/Bridge.lean`
Refined localization	`ScalarCertificates.lean`, `GlobalReduction.lean`
Coordinates and gauge	`GramAlignment.lean`, `CandidateAlignment.lean`, `Alignment.lean`
Actual Hessian and PSD	`BlockHessian.lean`, `Certificates/AmbientKernelAssembly.lean`, `Certificates/AmbientHessian.lean`
Local and global assembly	`LocalEstimate.lean`, `LocalCoercivity.lean`, `Endgame.lean`, `Completion.lean`
Additional stability	`QuantitativeStability.lean`

7.1 Supplemental bridge to the N7 project's N9 target

The publication-level `bridge/UpstreamN7Bridge.lean` imports the actual definitions from the pinned N7 challenge and proves its N9 stretch target, rather than a paraphrase of that target in new definitions. The translation uses $z = \sqrt{1 - r_*^2}$ and the explicit cap relabeling that sends upstream indices $(6, 7, 8)$ to candidate indices $(8, 6, 7)$. An explicit Lean audit confirmed that the new declaration `ThomsonN9.UpstreamN7Bridge.thomson_nine` has a type definitionally equal to the upstream N9 target. The isolated source compilation and audit completed with exit zero on October 6, 2026; the three reported axiom closures contain only the same standard axioms listed below.

The imported upstream challenge contains unproved placeholders, which produce expected warnings when that file is compiled in isolation. None is invoked by the bridge, and no placeholder

axiom appears in its proof dependency closure. The upstream source is obtained separately, not redistributed here. This is a supplemental Lean check: it is not a run of the separate Lean Comparator application, and the bridge was not included in the recorded nanoda export of the frozen N9 complete theorem. The publication-level `bridge/README.md` and `bridge/check-result.json` record the pins, source hashes, scope, and reproduction commands.

7.2 Finite artifacts

The exact data are inspectable independently of the numerical generators. The reduced polynomial and factored identity are in `near_sharp/reduced/near-sharp-d10-reduced-D60.json`; their literal Lean data and integer checks are in `lean/ClaudeL1/`. Its SHA-256 is

`9816ae66f3b68caf4a03d6c6f7c514124c8cd8ba4415497a79443ce64f158781`.

The corresponding reduced Bernstein and refined-window data are `near_sharp/reduced/bernstein-reduced.json` and `near_sharp/reduced/refined-windows-reduced.json`. Finite Gram tables are literal in `lean/ClaudeWindow/WindowData.lean`. The ambient local certificate is accompanied by `certificates/ambient-dd-k0.json`, while the selected Lean endpoint uses the literal interval circuit and congruence data in the `Certificates/Ambient*` modules. The source manifests bind all data; the JSON file is not an unchecked oracle imported at proof time.

Numerical SDP solvers, unrestricted numerical minimization, and certificate generation are discovery tools only. The separate exact Python replay command

```
python verify_global.py --reduced --refined
```

recomputes finite arithmetic premises. It is useful for independent checking, but a successful Python run is not equivalent to building the Lean proof. The authoritative, pinned Lean build and audit commands are in `REPRODUCE.md`; final theorem statements, axiom reports, and rejected negative controls must be checked, not merely the existence of object files.

7.3 Recorded checks and limitations

The baseline source-only verification recorded successful checks for all 181 selected project modules and 23 audit/control steps. Four false Lean mutations and a certificate coefficient corruption were rejected. The final axiom reports contain only `propext`, `Classical.choice`, and `Quot.sound`; there is no `sorryAx`, admitted theorem, or numerical oracle in the public theorem's dependency closure. Kernel-reduced finite reflection is used explicitly. The original build driver's metadata finalization was interrupted after the proof steps; a separate bound audit and finalization are disclosed in `verification/BUILD_FINALIZATION.md`. They are not represented as an observed uninterrupted driver exit.

The pinned environment is Lean 4.34.1 with Mathlib commit

`d13f23b723b8a846827a245b89c10fc7d3f11612`.

The second implementation, nanoda, accepted all 55,248 exported declarations in the dependency closure of `thomson_nine_complete`, printed the actual complete target, and permitted only the above three axioms. The checker is pinned to commit `3a2407216ee84a75f9e1aead6803d0578be06ae7`, and the exporter to `076e8e57707e813375e8f9da8bf989799ace9680`. The recorded run was on Windows. An earlier timeout remains an inconclusive attempt, not a proof rejection; the successful retry used the same exported proof. The optional POSIX recipe has not been executed. Full details and raw output are in `verification/secondary/README.md`.

Independent implementation checking reduces reliance on a single kernel, but still involves the exporter, implementations, compiler toolchains, and source-to-export binding. Neither it nor the primary Lean check replaces human examination of whether the statement captures the intended problem. No external referee's approval or independent operator's rebuild is claimed. The paper is a review draft; its explanatory text is not itself kernel checked.

Artifact availability. Version 0.1.0 repository: https://github.com/aravindkarthigeyan/thomson-n9-lean. Archive DOI: 10.5281/zenodo.23192843 (reserved during preparation; deposit and publication are not confirmed by reservation alone).

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References

[1] H. Cohn and A. Kumar, *Universally optimal distribution of points on spheres*, J. Amer. Math. Soc. **20** (2007), 99–148. arXiv:math/0607446.

[2] R. E. Schwartz, *The 5 Electron Case of Thomson's Problem*, Experimental Mathematics **22** (2013). arXiv:1001.3702.

[3] L. Kryvonos, L. Liehr, and M. A. Taylor, *Energy minimization for eight points on the sphere* (2026). arXiv:2609.22077v1.

[4] L. Liehr et al., *Energy-Minimization-8-Points*, source repository. https://github.com/lukasliehr/Energy-Minimization-8-Points. Inspected commit `50d14bc06bd41f61573eb6a36eedb8d00759af8f`. Also inspected: *Thomson-N-8-Warrant*, https://github.com/jstoobysmith/Thomson-N-8-Warrant, commit `2c35d62e07d72686931da4bc3e062d668e858eda`.

[5] H. Tran, *thomson-n7-lean*, paper and formal source (2026). https://github.com/huwngtran/thomson-n7-lean. Inspected commit `25f2fa53119273458cfbb3c4230904bffdd61e53`. The proved N7 endpoints and the separately unproved N9 challenge are distinguished in this source.

[6] *delsarte*, formal spherical-code and Schoenberg positivity library. https://github.com/specialjcg/delsarte. Vendored dependency commit `f0724e16b01d4561ee6cda8cd56db4db03de0496`; Apache-2.0, with compatibility changes documented in the artifact.

[7] *nanoda_lib*, independent Lean proof-term checker. https://github.com/ammkrn/nanoda_lib; commit `3a2407216ee84a75f9e1aead6803d0578be06ae7`. Proof export: https://github.com/leanprover/lean4export, commit `076e8e57707e813375e8f9da8bf989799ace9680`.